Retrofit Cost Analysis Report

Energy and Cost Summary

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database.

Building Information

- Weather File: USA_TX_Amarillo-Husband.Amarillo.Intl.AP.723630_TMYx.epw
- Building Name: SmallOffice
- Building Location: Amarillo, TX
- Total Floor Area: 511 m^2

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	**Wall upgrade:** Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation. **Roof upgrade:** The roof was upgraded to R-24.4 using Blown Fiberglass insulation. **Window upgrade:** • Upgraded to 3-pane glazing • Weatherstripping: silicone adhesive smoke gasket • Glazing film: low-e film • Window frame: wood window frame • Caulking: acrylic

		• Infiltration reduction: 30% **Door upgrade:** • Door type: wooden door • Bottom seal: automatic door bottom • Top/side seals: jamb weatherstrip • Infiltration reduction: 30%

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	195.48	189.95	**5.53**	**2.83%**

Key Cost Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr)

$-466 (-4.82%)

Scenario 1

Min Construction Cost Scenario

$9,000

Scenario 1

Cost Visualization

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison

Baseline $9,667

Best Saving Scenario $9,201

Cost Payback

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 19 yrs

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